

XU SI

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PROFESSIONAL EXPERIENCE

Georgia Institute of Technology, Atlanta, USA 2024 – Present
Postdoc in Geophysics (Supervisor: Zhigang Peng).

EDUCATION

University of Science and Technology of China (USTC), Hefei, China 2020 – 2024
Ph.D. in Geophysics (Supervisor: Xinming Wu).
China University of Geosciences (Beijing) (CUGB), Beijing, China 2017 – 2020
M.S. in Geophysics (Supervisor: Yijun Yuan).
China University of Geosciences (CUG), Wuhan, China 2013 – 2017
B.S. in Applied Geophysics.

RESEARCH INTERESTS

Earthquake detection and monitoring; Non-tectonic seismic event detection; Earthquake forecasting; Seismology foundation models; Geophysical artificial intelligence; Graph neural network; Large language models for geoscience.

PROFESSIONAL ACTIVITIES

Co-Convener

- INFORMS Annual Meeting, 2025 (Atlanta, GA)
- SSA Annual Meeting, 2025 (Baltimore, MD)
- Eastern Section SSA Annual Meeting, 2024 (Atlanta, GA)

Journal Reviewer Communications Earth & Environment; Journal of Geophysical Research; Surveys in Geophysics; Geophysics; Seismica; IEEE GRSL; Seismological Research Letters; BSSA; Patterns.

Field Work

- Seismic deployment: Summerville, South Carolina (2025)
- Seismic monitoring array deployment: Arabia Mountain, Georgia (2024–present)

MANUSCRIPT IN REVIEW

2. **Si, X.**, Z. Peng*, et al., Contrasting Aftershock Behaviors Following the 2023 Kahramanmaraş Earthquake Doublet.
1. **Si, X.**, X. Wu*, ..., Z. Peng, Tuning into Earth's Voice: Leveraging Large Language Models for Earthquake Detection, *In prepration*.

PUBLICATIONS

21. **Si, X.***, Z. Peng, Extremely Shallow Semi-Repeating Tremor Caused by Water Hammers in a Sewer Pipe in Social Circle, Georgia, *Seismological Research Letters*, 2026. <https://doi.org/10.1785/0220250421>.
20. Mach, P., **X. Si**, D. Wang, R. Shcherbakov, P. He, T. Taymaz, C. Ding, Z. Peng, Foreshock Behaviors and Mainshock Rupture Properties Associated with the 2025 Mw 8.8 Kamchatka Earthquake Sequence, *Earthquake Research Advances*, 2026. <https://doi.org/10.1016/j.eqrea.2026.100457>
19. Wang, H., S. Li*, F. Ping, **X. Si**, C. Zhang, Sea Ice Edge Constraint Improves Antarctic Sea Ice Seasonal Prediction in Deep Learning Models. *Adv. Atmos. Sci.*, 2026. <https://doi.org/10.1007/s00376-025-5024-2>

18. Si, X., X. Wu*, F. Hu, et al., Earthquake prediction in the Sichuan-Yunnan region based on graph neural networks and multi-modal data, *Chinese Science Bulletin*, 2025. <https://doi.org/10.1360/CSB-2025-0392>
17. Gao, H., X. Wu*, L. Liang, H. Sheng, X. Si, H. Gao, Y. Li, A foundation model empowered by a multi-modal prompt engine for universal seismic geobody interpretation across surveys, *Information Fusion*, 2025. <https://doi.org/10.1016/j.inffus.2025.103437>
16. Ma, S., Z. Li, D. Sun, Y. Su, J. Li, X. Si, J. Zhu, Global search of PKP precursors with graph neural network: Implications for scatterers in the lowermost mantle, *Geophysical Research Letters*, 2025. <https://doi.org/10.1029/2025GL115952>
15. Peng, Z., X. Lei, D. Wang, X. Si, P. Mach, Q. Zhong, C. Ding, Y. Deng, M. Qin, S. Miao, Mainshock rupture properties, aftershock activities and remotely triggered seismicity associated with the 2025 Mw7.7 Sagaing fault earthquake in Myanmar, *Earthquake Research Advances*, 2025. <https://doi.org/10.1016/j.eqrea.2025.100413>
14. Sheng, H., X. Wu*, X. Si, J. Li, S. Zhang, X. Duan, Seismic foundation model: A next generation deep-learning model in geophysics, *Geophysics*, 2025. <https://doi.org/10.1190/geo2024-0262.1>
13. Si, X., X. Wu*, Z. Li*, et al., An all-in-one seismic phase picking, location, and association network for multi-task multi-station earthquake monitoring, *Communications Earth & Environment*, 2024. <https://doi.org/10.1038/s43247-023-01188-4>
12. Si, X., X. Wu*, H. Sheng, et al., SeisCLIP: A seismology foundation model pre-trained by multi-modal data for multi-purpose seismic feature extraction, *IEEE Transactions on Geoscience and Remote Sensing*, 2024. <https://doi.org/10.1109/TGRS.2024.3354456>
11. Wang, S., X. Si*, Z. Cai, et al., Fast Global Self-Attention for Seismic Image Fault Identification, *IEEE Transactions on Geoscience and Remote Sensing*, 2024. <https://doi.org/10.1109/TGRS.2024.3436066>
10. Yang, Z., X. Wu*, X. Pang, H. Sheng, X. Si, G. Wang, L. Yang, C. Wang, Completing any borehole images, *IEEE Transactions on Geoscience and Remote Sensing*, 2024. <https://doi.org/10.1109/TGRS.2024.3469394>
9. Tan, M., X. Si, S. Teng, X. Wu*, X. Tao*, Comparative analysis of TPA-LSTM and transformer models for forecasting GEO radiation belt electron fluxes, *Space Weather*, 2024. <https://doi.org/10.1029/2024SW004119>
8. Dai, C., X. Si, X. Wu*, FlexLogNet: a flexible deep learning-based well-log completion method of adaptively using what you have to predict what you are missing, *Computers & Geosciences*, 2024. <https://doi.org/10.1016/j.cageo.2024.105666>
7. Jiang, L., X. Si, X. Wu*, Filling borehole image gaps with a partial convolution neural network, *Geophysics*, 2024. <https://doi.org/10.1190/geo2022-0344.1>
6. Wang, S., Z. Cai, X. Si, Y. Cui, A three-dimensional geological structure modeling framework and its application in machine learning, *Mathematical Geosciences*, 2023. <https://doi.org/10.1007/s11004-022-10027-9>
5. Wang, S., X. Si*, et al., Structural Augmentation in Seismic Data for Fault Prediction, *Applied Sciences*, 2022. <https://doi.org/10.3390/app12199796>
4. Zhang, Q., W. Zhang, X. Wu, J. Zhang, W. Kuang, X. Si, Deep learning for efficient microseismic location using source migration-based imaging, *Journal of Geophysical Research: Solid Earth*, 2022. <https://doi.org/10.1029/2021JB022649>
3. Yuan, Y., Y. Zheng, X. Si, Attenuation of linear noise using denoising CNN with asymmetric convolution blocks, *Exploration Geophysics*, 2022. <https://doi.org/10.1080/08123985.2021.1999772>
2. Yuan, Y.*, X. Si, et al., Ground roll attenuation using generative adversarial networks, *Geophysics*, 2020. <https://doi.org/10.1190/geo2019-0414.1>
1. Si, X., Y. Yuan*, et al., Attenuation of random noise using denoising convolutional neural networks, *Interpretation*, 2019. <https://doi.org/10.1190/INT-2018-0220.1>

INVITED TALKS

4. **International Conference on Earthquake Engineering (ICEE)**, Contrasting Aftershock Behaviors Following the 2023 Kahramanmaraş Earthquake Doublet Revealed by High-Resolution Catalog (July, 2026)
3. **Japan Geoscience Union Meeting (JpGU)**, Seismic Detection of Rock Exfoliation Events at Arabia Mountain, Georgia (May 25th, 2025)
2. **Georgia Institute of Technology**, School of Earth and Atmospheric Sciences, Graph neural network and foundation model for earthquake detection (September 20th, 2024)
1. **China Earthquake Networks Center**, Graph neural network for forecasting (January 5th, 2024)

SELECTED CONFERENCE ABSTRACTS

12. Conjugated Faulting during the July 25, 2023 Mw 5.5 Earthquake Sequence in Kozan, Southeastern Türkiye, P. Mach, Z. Peng, **X. Si**, O. S. Adeboboye, E. Zor, M. Ergin, & E. A. Sandvol
AGU Fall Meeting Abstracts, 2025.
11. Statistical Properties of Early Aftershocks Following Moderate-Size Earthquakes in the Central and Eastern United States, C. Ding, Z. Peng, J. Zhuang, **X. Si**
AGU Fall Meeting Abstracts, 2025.
10. Deep Learning-Based Forecasting of Induced Seismicity in Enhanced Geothermal Systems at Utah FORGE, **Si, X.**, O. E. Adeboboye, Z. Peng, J. Williams, W. Jin, & S. Dai
AGU–SEG Geothermal Workshop, June 2025.
9. Mechanistic Understanding of Induced Microseismicity during EGS Stimulation: A Utah FORGE Site Investigation, W. Jin, N. C. Acevedo, **X. Si**, S. Dai, Z. Peng, & J. Williams
AGU–SEG Geothermal Workshop, June 2025.
8. Seismic Detection of Rock Exfoliation Events at Arabia Mountain, Georgia (Invited Talk), **Si, X.**, Z. Peng, A. Reynolds, Y. Qian, P. Mach, C. Ding, K. A. Lang, M. C. Eppes, & S. Ni
JpGU Annual Meeting Abstracts, 2025.
7. High-Resolution Aftershock Catalog of the 2023 Kahramanmaraş Earthquake Sequence Reveals Detailed Fault Structures in Southeastern Türkiye, **Si, X.**, P. Mach, O. S. Adeboboye, H. Gao, E. A. Sandvol, E. Zor, M. Ergin, O. Yalvac, M. C. Tapirdamaz, A. Tarancioglu, F. Sevim, & Z. Peng
SSA Annual Meeting Abstracts, 2025.
6. Tuning into Earth’s Voice: Leveraging Large Language Models for Earthquake Detection, **Si, X.**, X. Wu, J. Li, X. Cui, H. Sheng, H. Gao, Z. Li, & Z. Peng
AGU Fall Meeting Abstracts, 2024, S11B-3384.
5. An All-in-One Seismic Phase Picking, Location, and Association Network for Multi-Task Multi-Station Earthquake Monitoring, **Si, X.**, X. Wu, Z. Li, S. Wang, & J. Zhu
AGU Fall Meeting Abstracts, 2023.
4. Earthquake Prediction in the Sichuan–Yunnan Region Based on Graph Neural Networks, **Si, X.**, X. Wu, F. Hu, et al.
Boao Disaster Prevention and Mitigation Conference, 2023.
3. Ground Roll Attenuation with Conditional Generative Adversarial Networks, **Si, X.**
SEG International Exposition and Annual Meeting, 2020.
2. Ground Roll Attenuation Based on Conditional and Cycle Generative Adversarial Networks, **Si, X.**, Y. Yuan, F. Ping, Y. Zheng, & L. Feng
SEG Workshop: Mathematical Geophysics, 2019.
1. Random Noise Attenuation Based on Residual Learning of Deep Convolutional Neural Networks, **Si, X.**, & Y. Yuan
SEG International Exposition and Annual Meeting, 2018.

HONORS AND AWARDS

Best Student Presentation Award, IPACES 2024 Annual Meeting	Jul. 2024
Best Student Presentation Award, 4th Workshop on Artificial Intelligence in Seismology	Jul. 2023

Best Student Presentation Award, Boao Disaster Prevention and Mitigation Conference	Feb. 2023
Outstanding Graduate, China University of Geosciences (Beijing)	Jun. 2020
Best Student Poster Award, 4th SEG International Workshop on Mathematical Geophysics	Nov. 2019
National Graduate Student Scholarship	Nov. 2019

TEACHING EXPERIENCE

EAS 8803: Advanced Data Analysis , Georgia Institute of Technology	Fall 2025
Guest Lecturer	
Delivered a 1-hour guest lecture on <i>Artificial Intelligence in Seismology</i> .	
GEPH 6401P: Artificial Intelligence in Geosciences , (Master's level), USTC	Fall 2021
Teaching Assistant	
ESS 1502: Artificial Intelligence in Geosciences , USTC	Fall 2020
Teaching Assistant	
ESS 1502: Artificial Intelligence in Geosciences , USTC	Fall 2020
Teaching Assistant	
Created and mentored the final project <i>Seismic Fault Segmentation Challenge</i> on Kaggle (https://www.kaggle.com/c/ustc-geo-ai20).	

MENTORING AND SUPERVISION

- **2025** Hullihen, Sarah (Georgia Institute of Technology, undergraduate summer research student): Head-wave detection of the 2025 Myanmar earthquake sequence (co-supervised with Z. Peng).
- **2020–Present** Co-supervised multiple undergraduate and graduate students on research in artificial intelligence for geoscience (author listed as a second author); details are provided in the *Publications* section.

SKILLS

- **Coding** - Python, Shell Scripting, and Latex. Basic knowledge of Matlab and C.
- **Softwares & Packages** - Pytorch, Pyg, NumPy/SciPy, Matplotlib, Pandas, Geopandas, TensorFlow/Keras, Sci-kit Learn, OpenCV, ObsPy, PyGMT, SAC, HuggingFace, lightning, Adobe Illustrator.
- **Computing** - Cloud Computing (AWS), Version control (Git).